

Education

University at Buffalo-Roswell Park Comprehensive Cancer Center | Buffalo, NY, USA

Ph.D. Cancer Sciences 2019 - 2025

- ★ Thesis focus: Immuno-oncology and targeted therapy

M.S. Cancer Sciences 2017 - 2019

- ★ Thesis focus: DNA damage response to small molecule anti-cancer agents

PSG College of Technology | Coimbatore, India

BTech. Biotechnology 2013 - 2017

- ★ Thesis focus: Molecular mechanisms of cell division

Experience

Research Associate (Machine Learning and bioinformatics)

Icahn School of Medicine at Mount Sinai, New York, NY (Remote)

Apr 2025 - present

- ★ Developing unsupervised machine learning workflows to identify cell of origin for plasma disease biomarkers in high-dimensional transcriptomic datasets.

Senior Research Aide (Pharmacodynamics simulation and modeling)

Enhanced Pharmacodynamics LLC, Buffalo, NY

Jan 2019 - May 2019

- ★ Developed an interactive web app using R Shiny, to predict immune toxicity of an anti-cancer drug candidate, based on existing population pharmacokinetic/pharmacodynamic (PK/PD) models.

Technical skills

Bioinformatics: R, Python, Linux/Command Line, Vibe-coding, RNA-seq, single cell RNA-seq, Perturb-seq

Molecular biology: Cloning, CRISPR-Cas9, PCR, primer design, cellular & animal models

Protein & Cell biology: Multiparametric flow cytometry (10+ colors), cell culture, immunoblot, proliferation assay

Publications

[Rajan, P., Zollo R., Guo Y., Alruwaili M., Zonneville, J., Lieberman M., Morreale B., Olejniczak S., Barbi J., Abrams, S., & Bakin, A. p38 blockade reverses the immune suppressive tumor microenvironment in metastatic breast cancer. 2025. Under review at NPI Precision Onc.](#)

[Alruwaili, M., Zonneville, J., Naranjo M., Serio, H., Melendy, T., Straubinger, R., Gillard, B., Foster, B.A., Rajan, P., Attwood K., Chatley, S., Iyer, R., Fountzilias, C., Bakin, A.V. A Synergistic Two-Drug Therapy Specifically Targets a DNA Repair Dysregulation That Occurs in p53-Deficient Colorectal and Pancreatic Cancers. Cell Reports Medicine, 2024, 5\(3\): p. 101434.](#)

[Luzhin A., Rajan P., Safina A., Leonova K., Stablewski A., Wang J., Robinson D., Isaeva N., Kanditze O., Gurova K. \(2023\). Comparison of cell response to chromatin and DNA damage. Nucleic Acids Research, 2023; gkad865.](#)

Awards & Honors

- ★ Completed Claude Science Hackathon (1 of 150 researchers selected worldwide)
- ★ Awarded for Academic Excellence by University at Buffalo
- ★ Supported by \$3000 grant award from University at Buffalo Mark Diamond Research Fund (UB-MDRF)
- ★ Selected as Career Experience Fellow by SUNY Research Foundation (SUNY-RF)
- ★ Selected for short-term project at Tata Institute of Fundamental Research (TIFR), Mumbai, India